Energy Economics 127 (2023) 107044

Contents lists available at ScienceDirect

Energy Economics

journal homepage: www.elsevier.com/locate/eneeco

Energy affordability across and within 26 European countries: Insights into the prevalence and depth of problems using microeconomic data

Micaela Antunes *, Carla Teotónio, Carlota Quintal, Rita Martins

University of Coimbra, CeBER, Faculty of Economics, Av Dias da Silva 165, 3004-512 Coimbra

ARTICLE INFO

JEL codes:
D63
L97
Q41

Keywords:
Energy affordability
Overshoot
Concentration index
Inequality
Multiple metrics
Europe

ABSTRACT

The main aim of this study is to assess energy affordability across and within European countries, using microeconomic data from the European Household Budget Survey.

Our results have shown great heterogeneity: the average energy burden ranges from 2.9% in Malta to 10.9% in Latvia, while the proportion of households with a burden above 10% is as low as 2% in Luxembourg and Malta, surpassing 43% in Hungary, Latvia and Slovakia. Affordability problems are concentrated among poorer households, as confirmed by negative concentration indexes in all countries analysed but are not restricted to the poorest - in 10 out of 25 countries, families in the second bottom income quintile also face problematic ratios. For families with affordability problems, the average burden might be extreme even in high-income countries like France, where it reaches 19%.

Beyond the inequality analysis, the comparison of different metrics uncovered issues that would go unnoticed under single-indicator approaches. Baltic and Central-Eastern European countries experience higher prevalence and depth of energy burden problems accompanied by high rates of arrears. Mediterranean countries like Cyprus and Portugal do not exhibit great energy burdens but struggle with inadequate thermal comfort, raising concerns about underconsumption. Bulgaria and Greece stand out with worrying results across all indicators. The panorama is less heterogeneous across Europe regarding efficiency-related problems in dwellings.

To be effective, measures should prioritise the most prevalent problems and consider their cross-cutting or group-restricted nature. In countries with more widespread and severe issues, solutions may require a co-financing by European Union. A common policy framework is needed to facilitate comprehensive and effective action, such as improving energy efficiency in housing stock and preventing the escalation of affordability problems due to the energy and climate crises, which disproportionately affect the most vulnerable households in Europe.

1. Introduction

From national governments and regulators to international organizations, there are multiple expressions of the recognition that energy affordability is a necessary condition to ensure universal access to energy services. One example is its inclusion in the United Nations 2030 Agenda for Sustainable Development, Sustainable Development Goal (SDG) 7 - Affordable and Clean Energy, with targets and indicators established to assess and monitor the progress of countries on the matter.

Energy affordability is sometimes used as a synonym and/or in complement to other concepts, such as fuel poverty or energy security. It emerges as one of the dimensions of the vaster phenomenon of energy poverty, although very often researchers and international organizations assess energy poverty through energy affordability. It is also related to the broader concept of vulnerable consumer. In Europe, the Electricity Markets Directive and the Gas Directive (EU, 2019a, 2019b) require Member States to ensure there are adequate safeguards to protect vulnerable consumers. The definition of vulnerable consumer is at the responsibility of Member States; some refer to the share of disposable income that consumers spend on energy, i.e. to affordability concerns; others, to energy poverty and the prohibition of disconnection from energy services to such consumers in critical times; and yet others to energy efficiency of homes or critical dependence on electrical equipment, for example due to health reasons.

Energy affordability was the cornerstone of the seminal research on

* Corresponding author.
E-mail addresses: micaela@fe.uc.pt (M. Antunes), carla.teotonio@fe.uc.pt (C. Teotónio), qcarlota@fe.uc.pt (C. Quintal), rvmartin@fe.uc.pt (R. Martins).

https://doi.org/10.1016/j.eneco.2023.107044
Received 12 August 2022; Received in revised form 10 September 2023; Accepted 15 September 2023
Available online 23 September 2023

energy/fuel poverty (see e.g. Isherwood and Hancock, 1979; Richardson, 1978). Historically, the term 'energy poverty' referred to inadequate energy access in less developed countries whilst the term 'fuel poverty' applied to energy affordability in developed countries. Nowadays these are used interchangeably (Castaño-Rosa et al., 2019; Das et al., 2022). In 1991, Boardman proposed a more formal definition of fuel poverty based on the 'energy burden', i.e., the percentage of household income spent on home energy services - in other words, the energy affordability ratio (EAR). According to this proposal, households spending more than 10% of their income to maintain comfortable indoor temperatures were considered to be in fuel poverty (Boardman, 1991). This has become widely known as the '10% indicator' and remained the only indicator for energy poverty for a certain time (Tirado-Herrero, 2017).

Energy services are usually considered to be affordable if the expenditures on them do not represent an excessive burden over households' income. However, there is no common definition nor a single measure to assess energy affordability, not to mention that the term 'excessive' relies on normative grounds. In the European Union (EU), for example, affordability issues are ultimately at the heart of the energy poverty problem, which, according to the EU Energy Poverty Observatory (EPOV) is understood to occur when energy expenditures represent a high percentage of consumers' income, affecting their ability to face other expenses, or when households are forced to reduce their energy consumption, thus affecting individuals' physical and mental health and well-being (EPOV, 2022). Energy affordability is, indeed, one of the four principles that European energy regulators should adhere (CEER, 2014).

Affordability issues affect significant shares of population, even in developed countries, particularly the most vulnerable consumers and contribute to explain the worrying estimates of people living in energy poverty, reaching around 34 million in Europe (EPOV, 2022), thus justifying the relevance of deepening the study of this phenomenon. Besides, affordability problems tend to increase in periods of downturns, which explains the adoption of financial and/or non-financial measures to support those most in need in Europe after the 2007/08 financial crisis.

With the economic recovery after the COVID-19 pandemic, and, shortly after, the war in Ukraine, Europe faces unprecedentedly high insecurity levels regarding energy supply and high energy prices, with expected impacts on energy affordability, and many consumers, especially the most vulnerable, being put under extraordinary strain. As a consequence, concerns about affordability of energy services are not expected to leave the European policy debate in the years to come.

Despite the criticality of the topic, the literature still lacks a comprehensive assessment of the energy affordability problems in the EU, which is essential to ensure inclusive growth. While acknowledging the relevance of looking at features such as low quality of housing stock, the ultimate goals of eradicating energy poverty or ensuring a fair energy transition for all cannot be achieved unless affordability issues are firstly resolved. On the other hand, regardless of which dimensions are analysed, it is not enough to investigate the prevalence of affordability problems. It is of utmost relevance to provide insight into their depth as well (Mulder et al., 2023).

Furthermore, the full understanding of the extent and the consequences of energy affordability problems, notably for specific groups of the population, calls for increased caution to its distributional aspects. In this regard, Frondel et al. (2015) highlighted the regressive impact of rising energy prices on the poorest households.

In short, a European-wide analysis of energy affordability is key for the design of the Community political framework to address this topic, as in the EU context most of the guidelines and strategies in this domain are primarily conceived at the supranational level. Such an integrated approach at the EU level, founded on a fine-grained dataset-based diagnosis of energy affordability issues across EU countries constitutes a fundamental basis for effective nationally-tailored policies and strategies to tackle affordability issues in each Member State.

In this context, the main aim of this paper is to assess energy affordability across and within European countries using microeconomic data and following an energy burden approach with emphasis put on distributive aspects. The second objective is to provide a holistic assessment of energy affordability in Europe, relying on different – yet complementary – approaches to energy affordability.

This paper contributes to the literature by providing an enriched analysis of energy affordability for a large sample of European countries, using microeconomic data. Studies for European countries are scarce and are very often based on national averages; however, averages can mask problems of affordability felt by some segments of the population. Hence, our contribution is further strengthened by an inequality analysis of energy affordability within and across countries, which we develop based on a range of methods. Another contribution to the literature comes from the development of a comparative analysis of affordability in the different countries, relying on four complementary indicators, to overcome potential limitations that may persist even from a distributive oriented analysis of affordability.

The paper is structured as follows. Section 2 provides a literature review on affordability studies, proposing an analytical framework to bring together the main indicators and metrics used. Section 3 describes the data and methodologies used in our analysis. In Section 4, results are presented and discussed. Section 5 concludes.

2. Literature review

In this section we start by proposing a taxonomy of measures to assess energy affordability. Then, we present some metrics that fit in each branch, highlighting their main advantages and weaknesses.

2.1. Taxonomy of energy affordability measures

Several metrics approach affordability in a direct/objective way, following the 'expenditure-based approach' usually found in literature (e.g. Thomson et al., 2017; Trinomics, 2016). Among these, the affordability ratios are the most popular. But notwithstanding the informative nature of affordability ratios, especially if using microeconomic data, some situations may not be properly captured unless other metrics are used in complement.

This explains why various indicators are sometimes used together to assess affordability problems (e.g., Deller et al., 2021; Dubois and Meier, 2016; Kyprianou et al., 2019). In particular, those metrics that allow to infer affordability in an indirect way through an assessment of households' (in)capacity to satisfy a set of socially perceived basic needs – thereby reflecting the affordability of energy services in an indirect way. This approach is, thus, similar to what is termed in several studies for Europe as 'consensual approach' (e.g. Castaño-Rosa et al., 2019; Thomson and Bouzarovski, 2018). It relies on subjective indicators collected from self-reported indicators of perceived conditions regarding thermal comfort and housing characteristics, whose lack is 'consensually' considered as a necessity by 95% of the EU population (Tirado-Herrero, 2017).

Based on this framework, this paper proposes a taxonomy for energy affordability measures anchored both on their direct or indirect nature and, further, on the perspective from which affordability is assessed (e.g., the burden viewpoint, the difficulties to make ends meet or the perceived conditions of thermal comfort). Table 1 summarizes these categories of measures and the main indicators used within each one.

2.2. Energy burden and other direct approaches

2.2.1. Energy burden and microeconomic data

The 'energy burden' category of measures is, among the direct approaches to assess energy affordability, the most used in developed countries (Berry, 2018; OECD, 2018). For that reason, the bulk of our analysis is based on this approach. In practice, these measures are

Table 1
Metrics to assess energy affordability.

Categories of measures		Indicators	Description
Direct approach	Energy burden	Energy affordability ratio	EAR over fixed thresholds EAR over relative thresholds
	Falling into income-poverty	Minimum Income Standard After Fuel Cost Poverty	Falling below poverty line after paying energy services
		Low Income High Cost	Falling below poverty line after paying energy services and simultaneously having an energy expenditure above the median
	Inability to pay	Arrears on utility bills	Inability to pay utility bills due to financial difficulties
Indirect approach	Perceived	Inability to keep home adequately warm	Perceived lack of financial capacity to keep comfortable indoor temperatures
		Dwelling comfortably warm during winter time	Perceived efficiency of the heating system and insulation conditions to keep the dwelling warm
		Dwelling comfortably cool during summer time	Perceived efficiency of the cooling system and insulation conditions to keep the dwelling cool
	Experienced	Presence of leak, damp, rot	Presence of leaks, damp or rot in the dwelling

Source: Own elaboration.

broadly based on affordability ratios between energy expenditure and income which are compared to thresholds that can be either absolute, i.e., a fixed percentage, or relative, i.e., considering the median or average weight of energy expenditure on income.

As to the absolute thresholds, the most commonly used is the '10% indicator' (also known as the Ten Percent Rule), meaning that households whose affordability ratio surpasses this limit are considered to have affordability problems. This threshold was defined in the 1990s (Boardman, 1991) considering approximately twice the median British household expenditure on energy by the poorest 30% of households and the energy cost necessary to keep living areas at 21° (Taylor, 1993). Nevertheless, more recently, other thresholds have been proposed. Colton (2011) considers a 6% threshold, based on the premise that housing costs should not exceed 30% of household income, and that household energy costs should not exceed 20% of housing costs. Drehobl et al. (2020) use the 6% threshold and further classify as high and severe energy burdens those exceeding 6% and 10% of households' income, respectively. Cook and Shah (2018) consider a 4% threshold instead. Households whose energy burden is greater than 4% but lower than 7% are considered as 'energy stressed'; households whose energy burden lays between 7% and 10% are 'energy burdened'; and households with an energy burden higher than 10% are 'energy impoverished'.

As to affordability ratios considering relative thresholds, the most common are the M/2 and 2 M indicators, both used by the EPOV to monitor energy poverty. The M/2 indicator, also called 'Hidden Energy Poverty' indicator (e.g. Castaño-Rosa et al., 2019; Kyprianou et al., 2019), presents the share of households whose energy expenditure (in absolute value) is lower than half the national median. Even if this could result from high energy efficiency standards, this metric allows for the identification of abnormally low expenditure levels for households whose income is too low and are thus forced to make trade-offs on essential goods to pay the utilities' bills (the so-called "heat or eat" phenomenon). The 2 M indicator presents the percentage of households whose share of energy expenditure in income is more than twice the national median share, thereby capturing households that allocate a remarkably high share of income to energy. This indicator does not allow, however, to distinguish between high expenses due to poor energy efficiency and high standards of living. Examples of relative thresholds include Agbim et al. (2020), who consider as 'objectively energy burdened' the Texan households who surpass an 8% threshold, and Berry (2018), who set a 9% threshold for French households.

Due to their relative easiness of operation, as well as their objectivity and potential to communicate, the 'energy burden' measures are a good starting point for affordability research (Das et al., 2022; Thomson et al., 2017).

Although widely used, some limitations of affordability ratios computed from effective expenditures are acknowledged. In particular, they do not consider an adequate energy consumption level, implying that they do not reveal possible underconsumption by poor households with (very) low consumption and thus low expenditure (Teotónio et al., 2023). Another limitation is that they may wrongly classify as affordability problems situations of overconsumption resulting from high living standards (Gawel and Bretschneider, 2014). Likewise, they do not consider some benchmark level of income that would impede to overstate affordability problems (high-income households can afford high energy costs without affordability problems coming from it (Rowley et al., 2015, Romero et al., 2018). Moreover, they disregard non-income factors that justify different consumption levels, such as household size and composition, health status, dwelling characteristics, technological endowments or climate conditions (Castaño-Rosa et al., 2019; Gawel and Bretschneider, 2014; Lin, 2018). These ratios are also very much sensitive to energy prices, and such aspect is frequently pointed out for leading to an under/overestimation of the problem of affordability when prices are low/high (Castaño-Rosa et al., 2019; Das et al., 2022; Romero et al., 2018). Nonetheless, energy prices are, in fact, a crucial aspect of affordability and their variability has very different impacts in higher-income households, who can accommodate such changes, vis-à-vis those with lower or stagnant incomes, who can be caught in a struggle to make ends meet (Meier et al., 2013).

In addition to the above limitations, others may remain with the use of aggregate data. And these apply not only to the energy burden measures, but also to the other direct measures described below, in subsection 2.2.2. Particularly in situations of strong inequality in income distribution, energy affordability assessments supported by aggregate data may inhibit a comprehensive understanding of the real dimension of the phenomenon. The relationship between energy expenditure and income can be traced back to Engel's Law of Consumption (1897), which shows that the share of energy expenditure in total expenditure is larger as income falls (Stigler, 1954). This finding remains valid nowadays, as shown by e.g. Simshauser (2021) for Australia and Das et al. (2022) for Canada. These latter authors further differentiate households in energy poverty, whose share of energy expenditure increases as income increases, and those not in energy poverty, to whom this share remains relatively unchanged across income levels. Indeed, it is a common finding in the empirical literature that low-income households spend a higher share of income on energy services than other income groups, despite that their consumption per capita is lower (Bagnoli and Bertoméu-Sánchez, 2022; Brown et al., 2020; Das et al., 2022). It is consensual in the literature that analyses supported by microeconomic data or based on income groups better capture differences in affordability at larger spatial scales (e.g. regions or countries), than averages, as these may hide the seriousness of affordability problems, notably those of low-income households (Goddard et al., 2021; Stiglitz, 2013). In this respect, it is even argued that within each income class, income should be considered discretionarily so as to avoid high ratios due to high/medium-high levels of income and associated high expenditure being confused with high ratios due to low incomes and expenditure levels that constitute a large burden in the households' budget (Goddard et al., 2021). Moreover, the discrepancy in the capacity to accommodating changes (rises) in energy prices between high and low-income households, who already record higher energy burdens, is noteworthy.

Although the importance of using microeconomic data from individual households is recognized (e.g. Deller, 2018; Simshauser, 2021), not so many examples exist in the literature, and, in addition, they are usually restricted to a single country or region (e.g., Agbim et al., 2020; Drescher and Janzen, 2021; Legendre and Ricci, 2015; Mulder et al., 2023; Oliveira-Panão, 2021).

2.2.2. Other direct approaches

This section briefly describes some direct measures of energy affordability found in literature, apart from the 'energy burden', notably those laying within the 'falling into income-poverty' and the 'inability to pay' categories (see Table 1).

The 'falling into income-poverty' category relies on the idea that households must face essential expenses beyond those in utilities (Deller and Waddams, 2015; Kessides et al., 2009) and thus compare the residual income (i.e., after deduction of housing and energy expenditure; see e.g. Trinomics, 2016) with a poverty line. Examples of these indicators include the 'Minimum Income Standard' (MIS), the 'After Fuel Cost Poverty' (AFCP) and the 'Low-Income High-Cost' (LIHC) indicators.

The MIS indicator considers different types of families (according to, e.g., household size, job status, disability) to set the corresponding Minimum Income Standard, i.e., the minimum income level that a household needs to be integrated in society (working as a poverty line) and then checks whether the residual income, i.e., net income deducted of housing costs and energy expenditure is below or equal to the respective MIS (Castaño-Rosa et al., 2019; Moore, 2012; Trinomics, 2016). The main advantage of this metric is its objectiveness in identifying energy affordability problems attributable to the lack of income, as it specifically identifies those households falling into poverty due to their energy bills (Romero et al., 2018). In turn, its main limitation relies on the determination of the MIS, which involves some arbitrariness, as it is highly relative and difficult to determine (Castaño-Rosa et al., 2019; Romero et al., 2018). To minimize this constraint, the AFCP adopts a threshold similar to the official poverty line, notably 60% of national median household income after housing and modelled fuel costs, further adjusted for household size and composition (Hills, 2012). This metric identifies those households who are not income poor a priori but are dragged into poverty by excessive fuel costs and, therefore, more likely to make trade-offs on essential goods. Nevertheless, it also labels as energy poor the very low-income households, irrespective of their fuel costs, thereby eventually mixing energy and income poverty situations that should be targeted by distinct types of policies (Hills, 2012).

Finally, the LIHC indicator compares i) the residual income after housing and energy costs with a threshold corresponding to the national official poverty line and ii) the required energy expenditure (i.e., the necessary cost to assure satisfaction of energy needs, given the housing conditions and household characteristics) with a threshold corresponding to the national median energy expenditure, adjusted for household composition (Castaño-Rosa et al., 2019; Martins et al., 2019; Thomson et al., 2017). Energy poverty (in practice, affordability issues) exists when, simultaneously, income is below the threshold and energy expenditure is above. This indicator has the advantage of considering an income threshold, thereby circumventing one limitation pointed to the energy affordability ratios (Romero et al., 2018). However, it does not capture the households who fall into poverty due to energy expenditure (Legendre and Ricci, 2015), as the previous two indicators, and, moreover, both the definition of the energy expenditure threshold and the 'doubly-relative' nature of this metric increase its complexity and decrease its transparency (Romero et al., 2018, p. 100).

The 'inability to pay' category relies on the indicator 'Arrears on utility bills' (also proposed by Thomson et al., 2017 as an affordability measure) since it provides objective and direct information on the segment of the population facing financial difficulties to pay utility bills on time. One main drawback is that it may underestimate problems as households may not have arrears due to restrictive behaviour, i.e., underconsumption. Another is the possible effect of different policies on such arrears in different countries, notably the existence of a service disconnection safeguard, which constitutes a caveat to the interpretation of this indicator associated with whether or not there is a service disconnection safeguard. In countries where such a measure exists, more households with arrears are expected to be found. This is one of the reasons why it is relevant to use different, yet complementary, metrics to analyse affordability to its full extent, especially when cross-section data is used. A third drawback comes from the fact that the 'Arrears' indicator accounts for other utility expenditures (Gouveia et al., 2022). However, considering that energy services usually represent a significant share of total expenditures on utilities (Cook and Shah, 2018; Yoon and Saurí, 2019), particularly for low-income households (Fankhauser and Tepic, 2007), it is reasonable to infer that families with arrears on utility bills are likely to be struggling to afford energy services (and, eventually, subject to disconnection of supply). Castaño-Rosa et al. (2019), Dubois and Meier (2016), Gouveia et al. (2022) and Thomson et al. (2017) use this indicator to build a composite indicator of energy affordability problems.

2.3. Indirect approaches

This section presents some of the most used metrics that assess affordability in an indirect way, in particular, indicators evaluating housing and household conditions like those obtained from the European Survey on Income and Living Conditions (EU-SILC).

The indicator 'Inability to keep home adequately warm' measures the share of households considering not have the financial capacity to keep homes satisfactorily warm. It is included in the Eurostat SDG7 indicator list and is used as a proxy for the affordability of energy services in the EU (European Union, 2021), although not capturing it perfectly (see, e.g., the empirical evidence in Agbim et al., 2020 and Deller et al., 2021). Moreover, many subjective factors condition this perception (e.g. age, sex, health status, climatological culture, standards or preferences) and, in addition, it only considers space heating, leaving behind other energy uses - two main drawbacks (Castaño-Rosa et al., 2019; Dubois and Meier, 2016).

This indicator also may correlate with the following two - 'Dwelling comfortably warm during winter time' and 'Dwelling comfortably cool during summer time' (Dubois and Meier, 2016). These report the share of the population considering that both the dwelling's insulation and heating/cooling systems assure that it is kept comfortably warm/cool during winter/summer. These are two of the indicators used by the EPOV to monitor energy poverty in the EU and contribute to assess households' affordability in the sense that the adequacy of indoor thermal comfort to climate conditions reveals, to a large extent, the affordability of energy services (in terms of heating/cooling systems or investments in energy efficiency). Still, these indicators are not included in the EPOV regular datasets (Gouveia et al., 2022). Concerning the latter, it properly accounts for the fact that energy poverty (and the associated affordability issues) is not limited to cold winters, even if summer energy poverty remains somehow overlooked in the EU (Thomson et al., 2017). Actually, one of the projected impacts of climate change is the increase in the frequency and intensity of heatwaves, which increases cooling needs and, therefore, affects energy demand and energy bills. For Europe, data show an increase in the cooling needs and a decrease in the heating needs since the 1980s - which are projected to continue or even accelerate - with the former effect particularly acute in Southern Europe, the Mediterranean region, and the Balkan countries, and the latter in Baltic and Scandinavian countries (EEA, 2017).

The indicator 'Presence of leak, damp or rot' measures the share of the population whose dwelling has a leaking roof, damp walls/floors/foundation or rotten window frames or floor. It directly assesses energy inefficiency and contributes, in an indirect way, to gain insight into energy affordability, given that households whose dwellings present these bad conditions are more likely to face not only high energy bills due to poor energy efficiency but also budget constraints that justify that poor housing conditions persist. The subjective character of this indicator is lower than that of the previous one (Gouveia et al., 2022) but, because it disregards income and households' characteristics it has to be combined with other indicators to be consistent (Castaño-Rosa et al., 2019).

The self-reported indicators are simpler to collect and to be used than the expenditure-based ones, but involve some subjectivity, either because i) different expectations or preferences make that judgments about the adequacy of income to satisfy energy needs differ between households' groups with identical characteristics, or ii) the embarrassment in recognizing the existence of these problems may produce misleading results (Agbim et al., 2020; Davillas et al., 2022; Thomson et al., 2017). In addition, these indicators in particular have a dichotomic structure, which hinders capturing different degrees of experience/perception (Thomson et al., 2017).

Given the limitations pointed out, an in-depth analysis of affordability may require using different indicators as complements, as already pointed out by Tirado-Herrero (2017). This is usual in energy poverty studies, justified by its multidimensional nature, but not so frequent when it comes to the affordability dimension alone.

In a recent study for the UK, Deller et al. (2021) concluded that the self-assessed inability to afford adequate warmth is much lower than the affordability problem captured by expenditure-based metrics (the 10% and LIHC indicators). Contrarily, Agbim et al. (2020) concluded, for Texas, that subjective measures (households unable or struggling to pay their electricity bills) captured a larger share of the population than objective measures (energy burden). They also concluded that the three metrics are associated, pointing towards that these indicators capture different ways of perceiving problems. In short, the differences found reflect the different dimension(s) each of them is measuring (Berry, 2018; Deller et al., 2021) and show they are rather complementary than substitutes.

Thus, using a sole indicator may inhibit a comprehensive understanding of the extent of (un)affordability. A complementary approach that combines distinct indicators can contribute to mitigating such incompleteness.

3. Data and methods

Data are taken from the 2015 wave of the European Household Budget Survey (EHBS), the most recent data available, released in 2021 (access was granted under the Eurostat research project RPP 60/2020-HBS). The EHBS is a micro dataset, with the information provided at the household level. For each country, the sample is representative of its population, thanks to the sample weights provided. We resort on a diversified - yet complementary - range of methods and consider 250,735 observations for 25 countries (BE-Belgium; BG-Bulgaria; CY-Cyprus; CZ-Czechia; DE-Germany; DK-Denmark; EE-Estonia; EL-Greece; ES-Spain; FI-Finland; FR-France; HR-Croatia; HU-Hungary; IE-Ireland; LT-Lithuania; LU-Luxemburg; LV-Latvia; MT-Malta; NL-The Netherlands; PL-Poland; PT-Portugal; RO-Romania; SE-Sweden; SI-Slovenia; SK-Slovakia).

The first step consisted of analysing expenditures on energy for each household *i*, Exp_energy_i, defined as expenses on electricity, piped gas, liquefied gas in cylinder, liquid fuels, coal, other solid fuels and thermal energy. Null expenditures were eliminated from the dataset because it was not possible to distinguish between 'true' zeroes and those that occur because energy bills are included in some other expenditure category, such as rents.

The Energy Affordability Ratio is computed for each household *i*, EAR_i, as indicated in eq. (1):

$$EAR_i = \frac{Exp\_energy_i}{Income_i} * 100 \tag{1}$$

where Exp_energy_i is the expenditure on energy of household *i* and $Income_i$ corresponds to the net income (total income from all sources including non-monetary components, minus income taxes) of household *i*. Afterwards, national averages (based on microeconomic data) are calculated, and, for each country, the average energy burden is also computed for different income quintiles. The next step consists of showing, for each country, the distribution of the EAR in graph form, plotting the cumulative percentage of households ranked by decreasing EAR on the horizontal axis and the EAR on the vertical axis. In this part of the analysis, it is also shown the share of households with an energy burden above the usual thresholds of 6% and 10%, considering such cases as indicative of affordability problems. Given that there is no consensus on the threshold beyond which one can peremptorily state that energy affordability problems exist, we compare the well-established, though tailored for the British past context, '10% indicator', with a threshold of 6% that assesses energy affordability problems on the wider array of the affordability of housing costs (see, e.g., Drehobl et al., 2020).

Let us represent by z the threshold for the EAR above which households are deemed to have affordability problems and define the variable $afford_i$ which equals 1 if $EAR_i > z$, and 0 otherwise. The total number of households, E, in each country, with affordability problems is given by eq. (2):

$$E = \sum_{i=1}^{N} afford_i \tag{2}$$

where N is population size. The share of the population with affordability problems, $\overline{afford}$, is then given by eq. (3):

$$\overline{afford} = \frac{E}{N} \tag{3}$$

This share captures the incidence of any affordability problem occurring but not its intensity. For that, we further calculate the overshoot (O'Donnell et al., 2008). The overshoot for household *i*, O_i, is given by eq. (4):

$$O_i = afford_i(EAR_i - z) \tag{4}$$

The mean positive overshoot (MPO) is then computed for each country, as shown in eq. (5) (O'Donnell et al., 2008):

$$MPO = \frac{1}{E}\sum_{i=1}^{N} O_i \tag{5}$$

The MPO shows, for each country, by how many percentage points the threshold is exceeded, on average, by households with affordability problems. As in the analysis of the share of the population with affordability problems, we compute the MPO both for the thresholds of 6% and 10%.

To analyse inequality in the distribution of affordability problems, we adopt the methodology of concentration indices (*CI*). Concentration indices measure relative inequality in one variable (in our case, the existence of affordability problems) over the distribution of another variable (in our case, net income of the household) and are often used to measure socioeconomic-related inequality. The *CI* is given by eq. (6) (O'Donnell et al., 2016):

$$CI = \frac{2cov(afford_i, R_i)}{\overline{afford}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{afford_i}{\overline{afford}}(2R_i - 1)\right] \tag{6}$$

where R_i is the fractional rank of household *i* in the net income distribution.

The convention is that the index takes a negative (positive) value when there is a disproportionate concentration of the variable (in this case, affordability problems) among the poor (rich). When there is no socioeconomic-related inequality, the concentration index is zero. The index is bounded between −1 and 1 (O'Donnell et al., 2008). Affordability problems are expected to be concentrated among poorer households, consequently, we expect to find negative *CI*. However, by computing *CI* it is possible to assess the magnitude of this concentration. We have estimated *CI* using the *conindex* STATA command written by O'Donnell et al. (2016). The statistical analysis was performed in STATA 15.1.

Because countries with the same prevalence of problems might have different degrees of relative inequality, and countries with the same relative inequality might have different levels regarding affordability problems, we show the position of countries in a two-dimensional diagram, with the prevalence of affordability problems on the vertical axis and the standard *CI* on the horizontal axis. A country might exhibit huge inequality but if we see that this is affecting a few households, then this tempers the concerns with affordability. Differently, it is worrisome when countries have simultaneously high prevalence and high inequality. Finally, if a country has a high prevalence and low inequality, then problems are relevant, but they are cross-cutting households of different levels of income (meaning that more universal measures might be required).

Finally, given the complementarity of the different metrics presented in Table 1, we assess and compare the prevalence of affordability problems in each country, resorting on four different indicators, related to the percentage of families: with EAR above 10%; unable to keep home adequately warm; with arrears on utility bills; and that report the presence of leak, damp, rot. Information for the last three indicators was collected directly from Eurostat (2022a, 2022b, 2022c). These indicators are computed by Eurostat based on the EU-SILC and encompass: i) the share of population not able to keep their home adequately warm, based on the question “Can your household afford to keep its home adequately warm?”; ii) the share of population having arrears on utility bills, based on the question “In the last twelve months, has the household been in arrears, i.e. has been unable to pay on time due to financial difficulties for utility bills (heating, electricity, gas, water, etc.) for the main dwelling?” and iii) the share of population with leak, damp or rot in their dwelling, based on the question “Do you have any of the following problems with your dwelling / accommodation?”. We retrieved information for the year 2015. Italy (IT) is not included in the analyses related to energy burden because its information on households' income is missing in the EHBS database. However, given that there is information for the indicators directly extracted from.

Eurostat, we include Italy in this comparative analysis. To this end, for households in Italy, the EAR is computed as the ratio between the expenditure on energy and total consumption expenditure of households (additional 15,013 observations from the EHBS).

To perform the comparative analysis, we represented the results in maps (created with ArcGIS 10.3), to facilitate the comparison across countries and/or indicators. A grey scale is used, with darker shades indicating higher prevalence of problems. To uniformize the scale across the four indicators, we follow Dubois and Meier (2016) procedure and, for each indicator, the score of 100% is attributed to the worst performance in terms of the prevalence of households with problems. For the remainder countries, the prevalence of problems is defined in proportion to the worst-performer country (for example, suppose that the country with the worst position in the arrears indicator has 30% of families with problems; then, 30% corresponds to the maximum value in the scale - 100% - and a country, say, with 15% of families with arrear problems appears with a value of 50%). This enables, on the one hand, to assess how better a country is relatively to the remainder, and, on the other hand, to map these indicators using the same amplitude for intervals.

4. Results and discussion

4.1. Level and distribution of households' energy burden

4.1.1. The energy affordability ratio (EAR) – Descriptive statistics

Table 2 presents the descriptive statistics regarding the EAR for the sample countries (ranked by decreasing order of the average EAR). The EU25 average EAR is 7.2%, ranging from 2.9% in Malta to 10.9% in Latvia. Six countries present an average EAR equal to or above 10%, while in 10 countries it is lower than 6%.

4.1.2. Decomposing the EAR: Average energy expenditure versus average income across countries

To better understand whether high (low) EAR mainly derive from high (low) energy expenditure or from low (high) income, we plot the variables that enter the computation of the EAR, in Fig. 1. On the horizontal axis there is the average annual net income per household and on the vertical axis, the average annual energy expenditure per household, for each country. The dashed lines indicate the average for each variable for the group of countries analysed.

The four quadrants that are visible from the dashed lines unveil different situations. Most countries with the highest EAR in Table 2 are in the bottom-left quadrant in Fig. 1, meaning that these ratios are mainly explained by low income. For example, Bulgaria and Romania are the countries with the lowest absolute expenditures and still present high EAR. For countries with the lowest EAR (bottom of Table 2), we observe varied situations. Malta shows the lowest EAR but this is not due to a particularly high net income. Finland, Luxembourg, and Sweden, also have low EAR, up to 4%, but are in different positions compared with Malta, given that these countries have higher energy expenditure accompanied by higher income, especially, Luxembourg, which has the second highest energy expenditure in the sample and still has the third lowest EAR. Denmark appears isolated in Fig. 1. Although this country has the second highest income in the sample, its exceptionally high energy expenditure (the highest in the sample) puts it in the middle of the ranking in Table 2.

Table 2
Energy affordability ratio (unit: %): descriptive statistics.

Country	Average	95% Confidence Interval		Min	Max	# Obs
LV	10.9	10.6	11.2	0.3	88.9	3841
HU	10.6	10.4	10.8	0.2	87.8	7180
SK	10.6	10.4	10.8	0.2	83.9	4635
BG	10.4	10.1	10.6	0.2	75.9	2961
CZ	10.1	9.9	10.4	0.0	65.7	2846
HR	10.0	9.7	10.2	0.2	58.5	2018
PL	9.8	9.7	9.9	0.0	99.4	35,069
RO	9.5	9.4	9.6	0.2	98.5	29,953
LT	9.2	9.0	9.5	0.4	77.1	3427
EE	8.7	8.4	9.0	0.1	91.0	3274
SI	8.4	8.1	8.7	0.8	86.1	3748
EL	7.8	7.6	8.0	0.2	94.6	6118
PT	7.0	6.8	7.1	0.1	98.9	11,352
DK	6.7	6.3	7.1	0.0	72.8	2188
DE	6.5	6.4	6.5	0.0	95.5	50,883
NL	5.7	5.5	5.8	0.2	86.9	14,377
BE	5.5	5.4	5.6	0.0	92.9	6131
IE	5.0	4.8	5.1	0.0	86.1	6673
ES	4.7	4.7	4.8	0.0	82.2	21,800
FR	4.2	4.2	4.3	0.0	100.0	16,586
CY	4.2	4.1	4.3	0.0	30.3	2856
SE	4.1	3.9	4.4	0.2	67.7	2469
LU	3.5	3.4	3.6	0.1	61.6	3166
FI	3.1	3.0	3.2	0.0	79.3	3670
MT	2.9	2.8	3.0	0.0	57.9	3514
EU25	7.2			0.0	100.0	250,735

Note: Null energy expenditures were eliminated from the dataset.

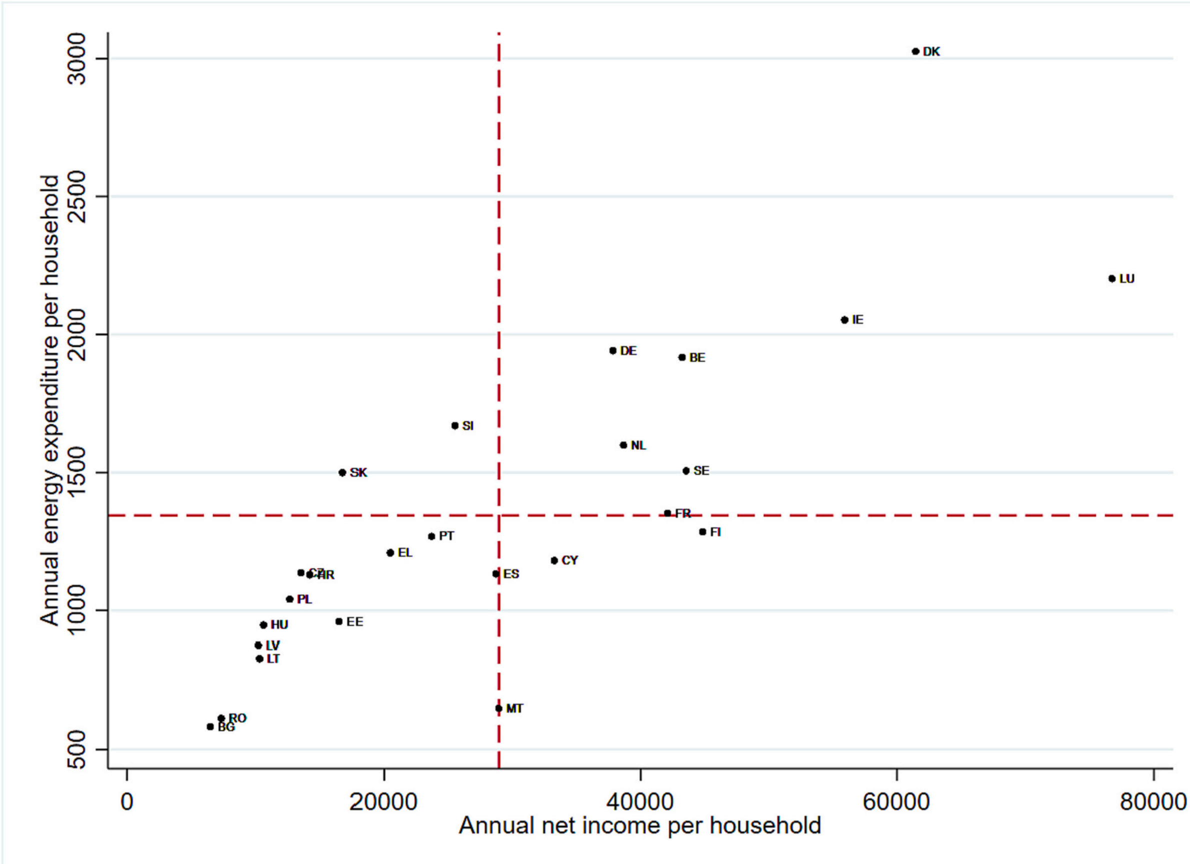


Fig. 1. Scatterplot for annual energy expenditure per household versus annual net income per household.

4.1.3. The energy affordability ratio (EAR) across income quintiles

Because affordability ratios might vary depending on the groups of households considered, Fig. 2 shows EAR by country and income quintiles, ranked as in Table 2.

As shown in Fig. 2, EAR greatly vary across income quintiles, in most countries. As expected, in all countries the average burden of energy expenditure decreases as we move from the first to the fifth quintile though the gradient varies across countries. In 15 countries, the poorest

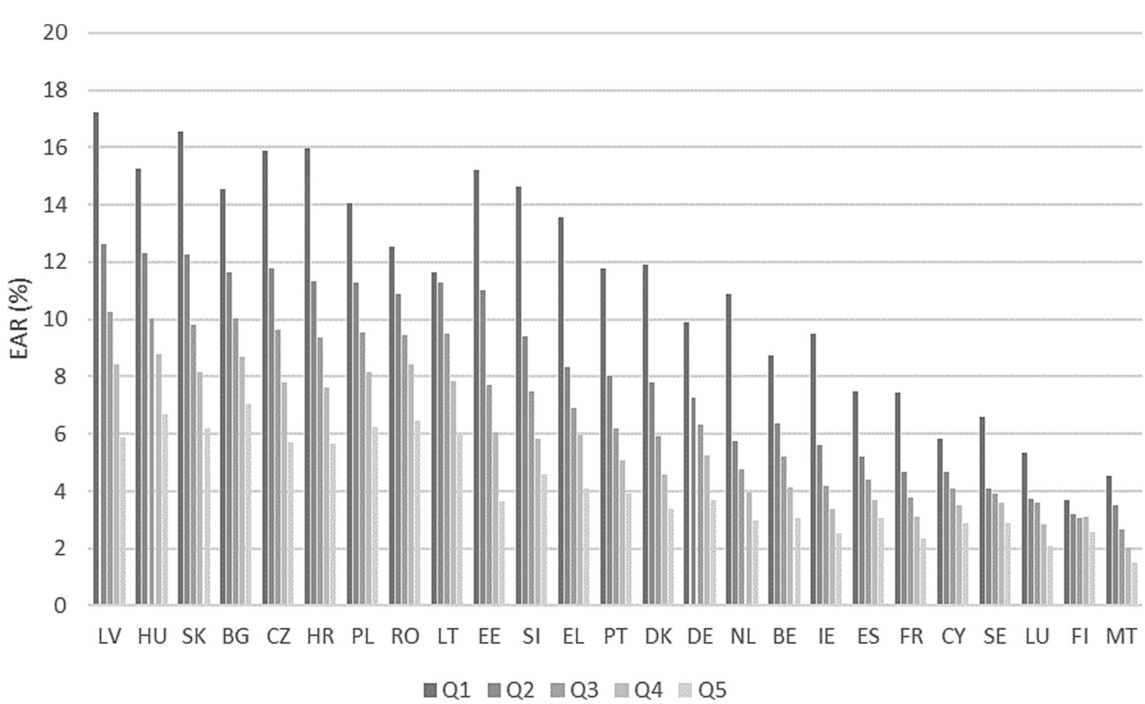


Fig. 2. The Energy Affordability Ratio (EAR) by income quintiles.

spend on average more than 10% of their income on energy. In 10 of these countries, even for the second bottom income quintile, the average EAR surpasses 10%. In some countries, the difference between ratios in the first- and second quintiles is quite large. This happens mostly in countries with high average EAR in Table 2 but it also occurs in countries with low average EAR. An astonishing example is that of The Netherlands, a country with an EAR of 5.7% in Table 2, though showing an EAR for the poorest above 10%, in Fig. 2. In general, the highest average EAR among the poorest households are found in countries with the highest average EAR in Table 2. Nonetheless, it is visible in Fig. 2 that the ranking of countries does not fully coincide with the ranking that would apply if based on the EAR of the poorest.

4.1.4. Distribution of the energy affordability ratio across households and the share of the population exceeding the thresholds of 6% and 10%, by country

Looking beyond averages, in Fig. 3 (countries appear in the same order as in Table 2), one might observe the distribution of individual EAR in each country and what are the percentages of households spending on energy more than 10% and more than 6% of their income.

Fig. 3. (Continued) Distribution of the EAR and % of households with EAR>6% and EAR>10% ($\overline{afford}$), by country.

Note: In the vertical axis it is indicated the households' EAR.

The percentages of households with EAR higher than 10% vary from about 2% in Luxembourg and Malta to more than 40% in Hungary, Latvia and Slovakia. The ranking of countries based on the average EAR is not exactly the same as the ranking based on the percentage of households with EAR above the thresholds considered. Nonetheless, there is an overall match between the two orderings, meaning that the general tendency is that countries with the highest average EAR are those with the highest prevalences of problems. Still, Fig. 3 brings into the light differences not visible in Table 2. For example, Belgium and Germany have an average EAR of 5.5% and 6.5%, but they have almost 10% and more than 16% of families with EAR above 10%, respectively.

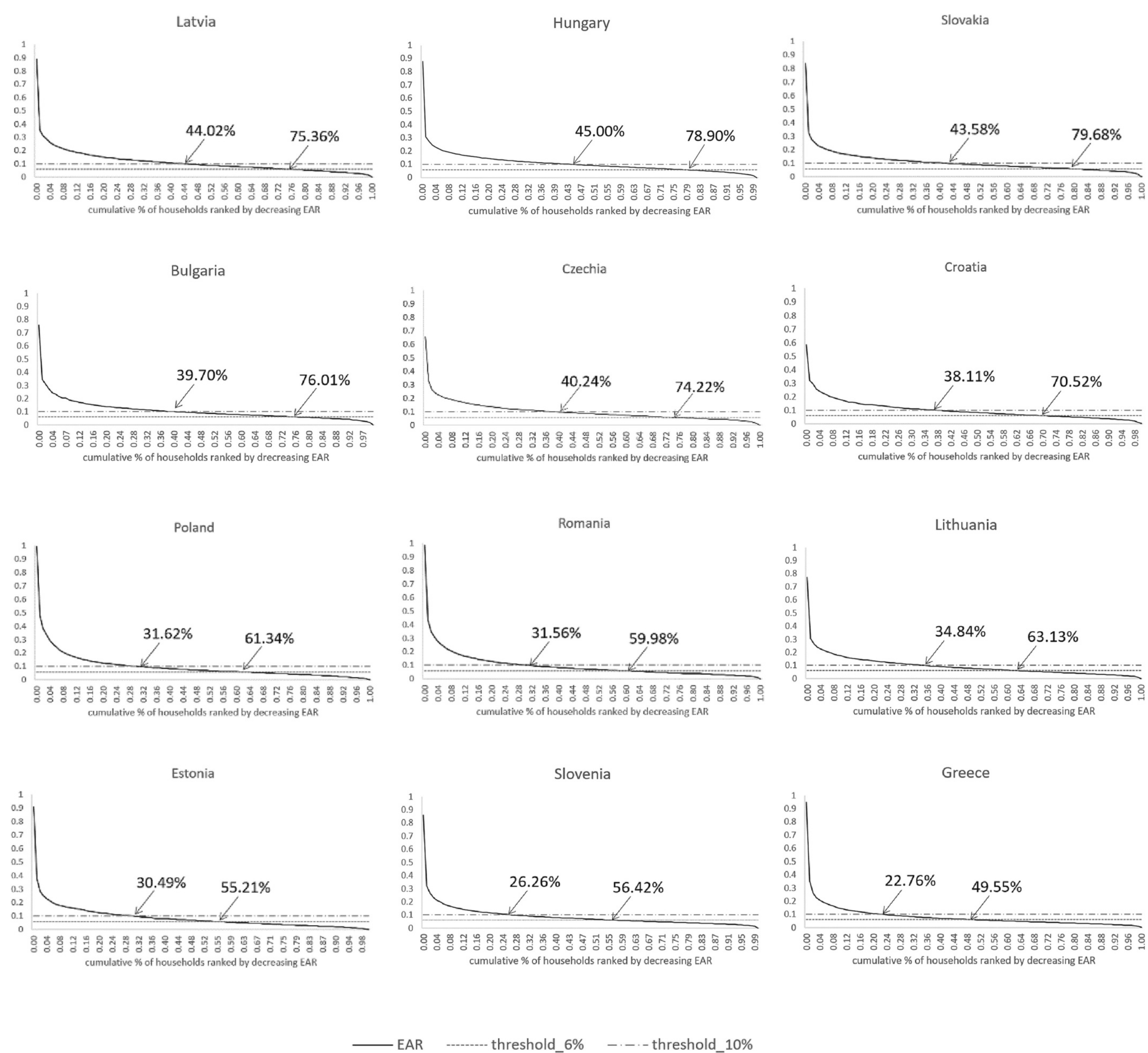


Fig. 3. Distribution of the EAR and % of households with EAR>6% and EAR>10% ($\overline{afford}$), by country.
Note: In the vertical axis it is indicated the households' EAR.

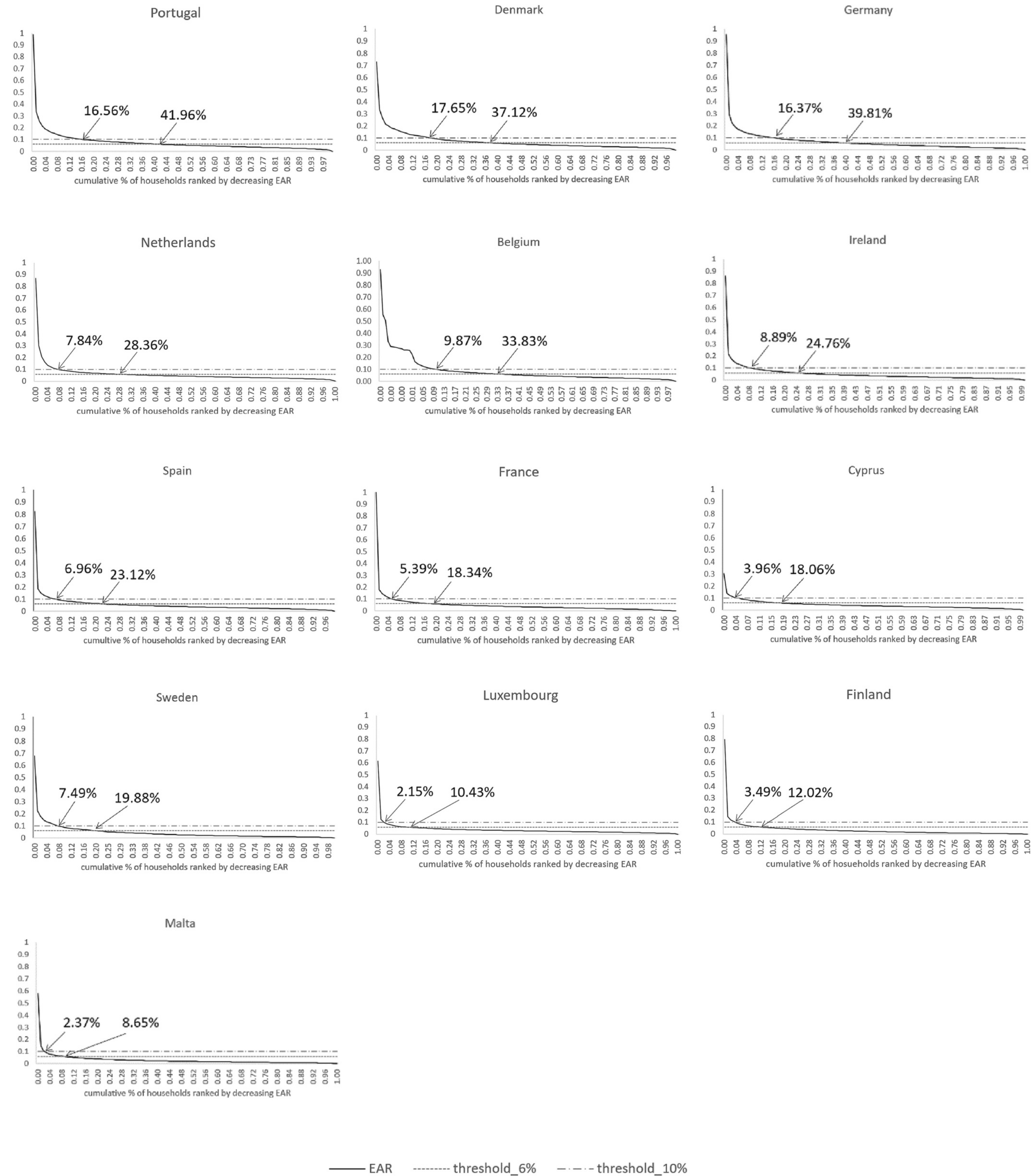


Fig. 3. (*continued*).

Considering the threshold of 6%, four countries have three-quarters or more of households with affordability issues.

In addition to the percentage of households with EAR above 6% or 10%, Fig. 3 shows the overshoot, that is, the extent to which families with problems exceed the thresholds. Graphically, the overshoot corresponds to the area under the curve but above the threshold level (10%, or 6%). In some cases, like France, the overshoot is concentrated among a few families. In other cases, like. Croatia, the overshoot is spread among a larger percentage of households. Based on Fig. 3, the overshoot seems to be the lowest in Cyprus. In the next subsection, the overshoot is quantified.

4.1.5. Mean positive overshoot

Fig. 4 shows, by country and for those families with affordability problems, by how many percentage points they exceed, on average, the reference thresholds. The countries are ranked according to the decreasing order of average EAR, as in Table 2.

Taking into account the threshold of 10%, four countries stand out in Fig. 4. The case of Poland is striking, given that families with affordability problems spend on energy, on average, almost 20% of their income. The case of France is also worth mentioning as this country has a relatively low EAR and low share of households with affordability problems, however, for the affected families, their average EAR is about 19%. Considering the 6% threshold instead, in general, the MPO is lower as larger shares of households are at stake.

4.1.6. Prevalence of affordability problems and relative inequality

The two previous sections highlight the prevalence and depth of affordability problems, but they do not show how these problems are distributed across households. Thus, Fig. 5 combines information on the representativeness of affordability issues (percentage of families with EAR above 10%) with the distribution of these problems (shown by the *CI*). As expected, in all countries, the *CI* is negative, meaning that affordability problems are disproportionately concentrated among poorer households.

In general, countries with a lower prevalence of affordability problems show greater concentration (bottom-left quadrant), meaning that the relatively fewer cases of affordability problems occur mainly among the poorest. Differently, the top-right quadrant in Fig. 5 is populated with countries from the top of Table 2 (highest EAR), which are also countries with a higher prevalence of affordability problems. Still, Fig. 5 discloses some differences - in the group of countries with prevalence of affordability problems ranging from 30% to 45%, the *CI* still ranges from −0.464 (Estonia) to −0.201 (Romania). Fig. 5 complements previous analyses as it shows differences that would go unnoticed without the inequality assessment. However, Fig. 5 also shows seemingly equivalent situations, which stem from quite different contexts. This is the case of Denmark and Portugal, which are close in this relative analysis as well as in Table 2. Nonetheless, in Fig. 1, where absolute values for average income and energy expenditure are depicted, they are quite apart from each other. When replicating the analysis for the threshold of 6% (available upon request), there is a clearer separation between countries on the bottom-left quadrant, on the one hand, and countries on the top-right quadrant, on the other. Finland and Sweden stand out with much lower inequality, compared to countries with similar prevalence of affordability problems.

4.2. Using different metrics to assess affordability: A comparative analysis

The relative positions of countries regarding four complementary indicators of energy affordability are shown in Fig. 6. Lighter shades mean that the countries are in a better position than those with darker shades, but that does not necessarily mean the absence of affordability problems. Some countries consistently appear in lighter or darker shades for all indicators, while others change depending on the indicator.

Note: The score 100% (worst performance) corresponds to: in panel (6a), 45% of households with EAR above 10% (Hungary); in panel (6b), 42% of households with arrears on utility bills (Greece); in panel (6c), 39.2% of households unable to keep home warm (Bulgaria); in panel (6d), 28.1% of households with leaks, damp or rot in the dwelling (Portugal).

How to read: In panel (6a), for example, the score of 100% is attributed to the highest value of this indicator, which is 45% (Hungary). In Latvia, there are 44% of households with EAR above 10%, then, the score 97.7% is attributed to this country. This means that Hungary and Latvia appear with the same shade, in this case, the darkest shade (band 80–100). Lithuania and Poland have 34.8% and 31.6% of families with EAR above 10%, then, the scores 77% and 70% are attributed to these countries, respectively. Both appear with the same shade (the second darkest), corresponding to the band 60–80. And so forth.

Source: Own calculations from European Household Budget Survey and Eurostat (2022a, 2022b, 2022c) data.

Notwithstanding the heterogeneity found across countries, one might say that two countries, Bulgaria and Greece, present generally unfavourable results in all indicators, while Finland and Sweden are in the opposite situation. Other countries not showing a significantly high prevalence of problems in any of the indicators analysed are Luxembourg, The Netherlands, France, Belgium, Denmark, Spain, Ireland and Malta. A different group of countries comprising Czechia, Slovakia, Estonia, Poland, Croatia and Romania presents as its main feature high shares of households with EAR above 10%. A further group, including Italy, Hungary, Latvia, Slovenia and Lithuania, is also

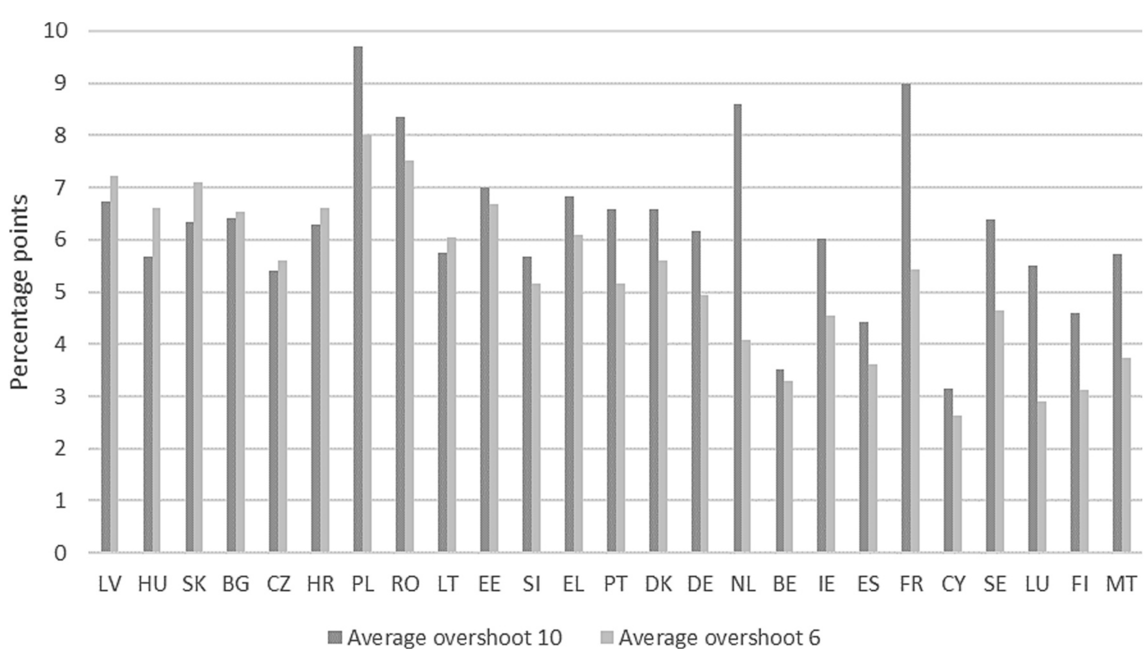


Fig. 4. Mean Positive Overshoot (MPO), by country.

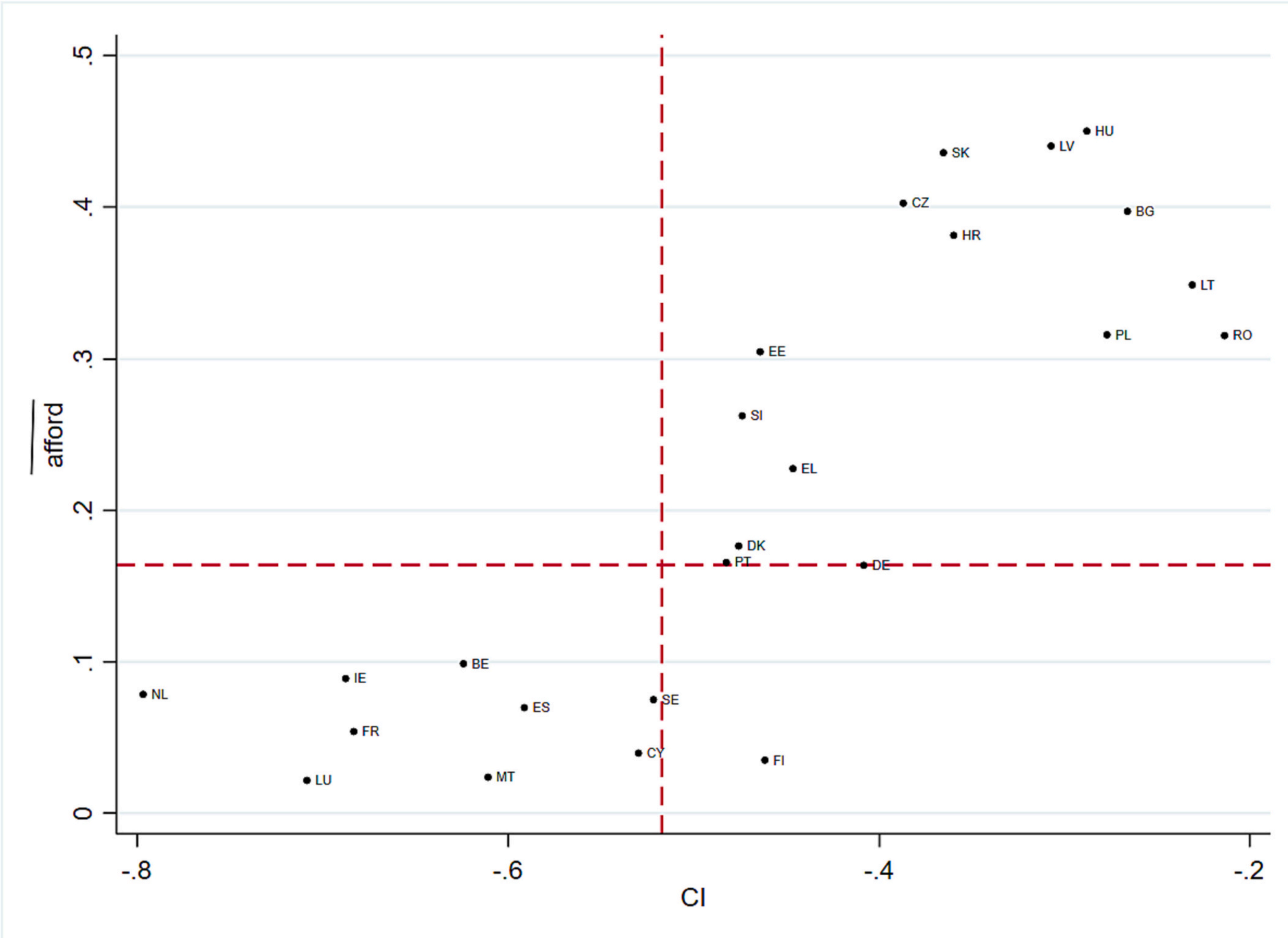


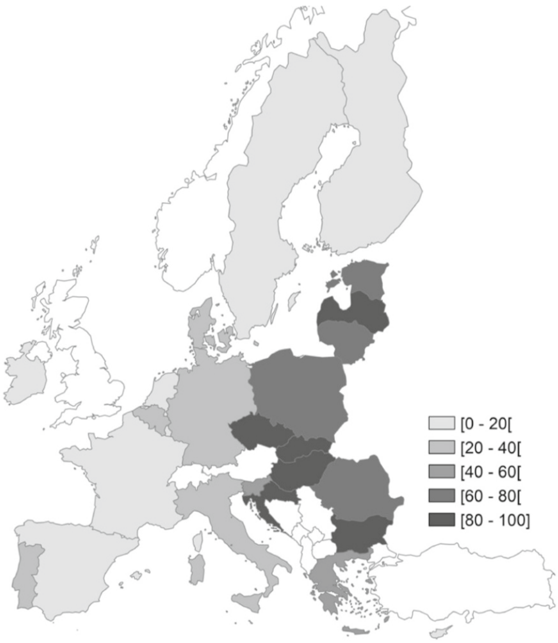
Fig. 5. Level and income-related inequality in affordability problems (threshold 10%).
Notes: $\overline{afford}$- prevalence of affordability problems; *CI* - Concentration Index.
All *CI* are statistically significant at 1%.

characterised by high shares of households with EAR above 10% but now accompanied by a greater prevalence of problems in the remainder indicators. Finally, Portugal and Cyprus differ from the previous two groups because they exhibit a considerably lower proportion of households with EAR above 10%; however, they are among the countries with the five highest rates of prevalence regarding problems with lack of thermal comfort.

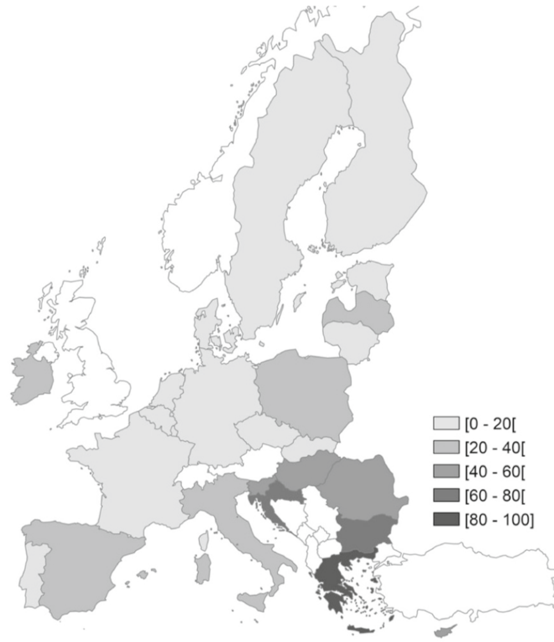
Comparing the different approaches, one can conclude that in some cases the EAR indicator is enough to show the occurrence, or otherwise, of affordability problems. Nonetheless, in other countries, where high burdens are combined with a high prevalence of the other indicators, the EAR signals the occurrence of affordability issues, but it does not show their full extent. Finally, in some countries, the EAR does not signal any problem at first glance, but it may be completely overlooking the other dimensions/problems here identified. It is quite a revealing fact that, in all Southern European countries, the indicator with the highest prevalence is 'inability to keep home adequately warm'. This means that their relatively low EAR might be concealing a problem of under-consumption, either because homes are not equipped with e.g. central heating or simply because households avoid using their equipment to save on energy costs (Dubois and Meier, 2016).

But, more than any geographically-, or climate-, related grouping of countries, it seems that, in general, countries that appear in Fig. 1 above the average income, not only appear with the lowest ratios in the bottom half of Table 2 but also present lighter shades in Fig. 6 (leaving panel (6d) aside, given that it shows a more homogeneous panorama across Europe regarding physical conditions of dwellings). This means that, regardless of the level of expenditures on energy, income seems to be key in preventing high energy burdens as well as in preventing under-consumption and arrears. Still, despite the apparent pivotal role of income, energy affordability problems might be mitigated through varied measures oriented to lower expenditure.

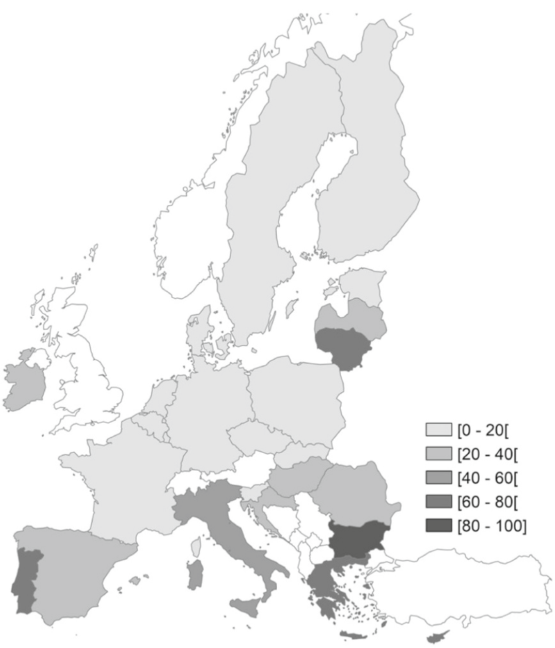
These measures are the more critical as our analysis has shown that, even within high-income countries, affordability problems can reach alarming levels among the poorest. The cross-country analysis presented in this work is pertinent, above all, to emphasise the complementary nature of different approaches to energy affordability and to signal which type of problems might be more relevant in each country. Nonetheless, it is also of utmost importance to pay attention to variations within countries. In addition to the focus on the prevalence of affordability problems, in this study, we have also analysed the distributional dimension of the energy burden. Not surprisingly, in countries with high average EAR, the poorest suffer the most. But this is also true in countries with lower EAR. While acknowledging that the most serious affordability problems occur in countries with average income below average, in countries with low EAR, such as The Netherlands and France, the mean positive overshoot is quite high. In these countries, fewer households face energy affordability problems, but those who do are severely hit. We also found that affordability problems are quite concentrated among the poorest in those two countries as well as in



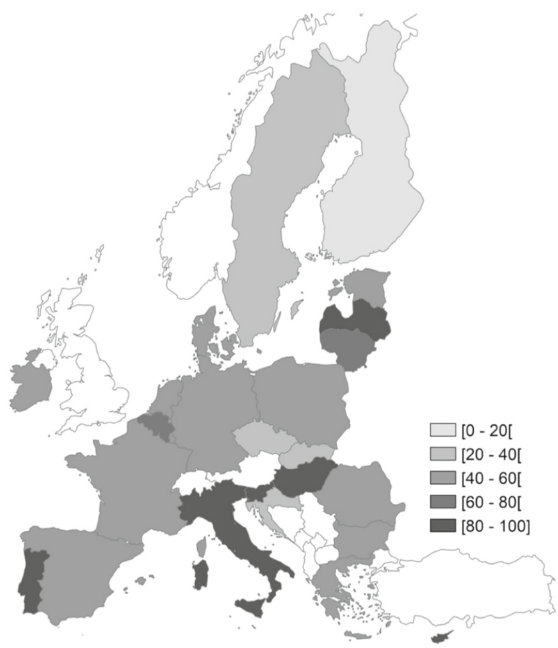
(6a) Based on the indicator: '% of households with EAR > 10%'



(6b) Based on the indicator: '% of households with arrears'



(6c) Based on the indicator: '% of households unable to keep home warm'



(6d) Based on the indicator: '% of households with leaks/damp/rot in dwelling'

Fig. 6. Distance to the worst performer country regarding the percentage of households with: EAR above 10% - panel (6a); arrears on utility bills - panel (6b); unable to keep home adequately warm - panel (6c); with leaks, damp or rot in the dwelling - panel (6d).

Luxembourg, which is the country with the highest average income in the sample. In this regard, it should be noted that there are concerns that the CO_2 tax, in Luxembourg, will affect the most vulnerable who tend to live in, and use, less efficient buildings and domestic appliances (Baptista and Marlier, 2020). This is actually a general concern, as households with high energy burdens are more likely to stay caught in cycles of poverty (Bohr and McCreery, 2019). Some indicators, like LIHC, have been used to identify low-income households in inefficient dwellings that require energy efficiency improvements. However, the efficacy of measures that have been conceived to tackle these problems is questionable in terms of affordability improvements (Berry, 2018; Castaño-Rosa et al., 2019), as they hardly benefit those most in need due to illiteracy, bureaucratic or financial issues (Baptista and Marlier, 2020; European Commission, 2019).

There are other examples of measures targeting the most vulnerable. In Finland, for instance, low-income households are supported by the (national) basic social assistance and municipal supplementary and preventive social assistance, to pay their electricity bills or other energy-related costs. In Sweden, access to energy is covered by social assistance for costs related to housing. In Portugal, France or Greece, the eligibility criteria for social tariffs are based on a series of social welfare criteria, i.e. benefiting from certain social transfers also entitles the holder to a social energy tariff, as highlighted by Dobbins et al. (2016). Other types of assistance to low-income households may include in-kind benefits like the provision of wood and coal by municipalities in Hungary (Baptista and Marlier, 2020). In the case of Malta, although one cannot exclude some potential problems of underconsumption, this is the country with the lowest average EAR and the lowest proportion of families with EAR above 10% in the sample, but its average income is 'only' about the sample average. What happens in Malta is that low-income households may be entitled to an amount to offset 30% of the consumption of electricity, as well as to an annual gas rebate per household and a subsidy for the rental of water and electricity meters (Baptista and Marlier, 2020).

There are nonetheless some limitations applying to our analysis worth mentioning. One drawback may come from the fact that null energy expenditures were eliminated from the dataset, for being impossible to distinguish between 'true' zero expenditures and those that meant the inclusion of energy expenditures on other expenditure categories. Still, this has a residual expression in the dataset, representing about 2% of the EU25 families.

In panel (6c), although the focus is on problems to keep homes warm, climate change and rising temperatures are likely to bring the opposite situation - problems to keep the dwelling cool. The EPOV uses as primary indicators of energy poverty, not only 'Inability to keep home adequately warm' but also 'Inability to keep home adequately cool during summer'. However, there are problems with the availability of data on the latter (Castaño-Rosa et al., 2019; Thomson and Bouzarovski, 2018). Existing evidence is still scarce and context-specific, like the survey carried out in 2016, in Maricopa County, Arizona (Maricopa County Department of Public Health Office of Epidemiology, 2016). This study found that one-third of individuals faced limitations on home cooling system use, with the majority (81%) citing the "cost of bills" as a contributing factor. Sooner rather than later, surveys like the EHBS should incorporate concerns with the impacts of climate change projected for Europe. While the decrease in heating needs projected for the Nordic countries may, ceteris paribus, lighten the energy bills and thus contribute to even lower energy burdens in these countries (note that these countries broadly record EAR below the 6% threshold), the projected increase in cooling needs in the Southern countries will worsen the average energy burden of their households, which is already exceeding 6% in some countries and even 10% in others.

Moreover, energy needs go beyond heating or cooling. They encompass the energy required to maintain health and participate in a productive economic and social life, being defined with reference to a decent standard of living in a particular place (Fournier et al., 2020). Even in the case of the well-established approach of the energy burden, the '10% threshold', adopted in the 1990s considering the UK context, may nowadays be considered unsuitable (Castaño-Rosa et al., 2019). Also, for high-income countries, spending 10% of income on energy might be bearable given that the income left is plenty enough to make ends meet, while in other countries spending on energy even less than 10% of income might force families to make difficult "heat or eat" choices (European Commission, Directorate-General for Employment, Social Affairs and Inclusion, et al., 2020). In addition, within countries, national thresholds may not account for spatial differences in energy needs (Agbim et al., 2020).

Regarding the arrears indicator, in panel 6(d), one must be cautious in the interpretation of results because the percentage of households reporting this problem might be higher in countries where there are measures to protect low-income households against disconnection from the energy supply, which according to Dobbins et al. (2016) vary across EU Member States. Nonetheless, bans on proceeding with disconnection are usually restricted to delayed disconnection or limited to specific calendar periods. In Greece, the country with the highest percentage of households reporting arrears (42%), bans on disconnections apply from November to March and in July and August. But the prevalence of arrears might be high in countries without such bans. In Hungary, for instance, almost a fifth of households (19.4%) have reported arrears though there are no restrictions to energy disconnections. What happens in this country is that families in arrears must be informed of the options available (that is, support measures and respective eligibility conditions) so as to avoid service disconnection (Baptista and Marlier, 2020).

The data used in this study was collected some years ago, despite its recent release, in 2021. In the meantime, the world has witnessed some severe shocks, such as the COVID-19 pandemic and rising energy costs, which will exacerbate the problems here diagnosed and the saliency of our findings (Drehobl et al., 2020). These energy shocks are also likely to invert the seemingly favourable evolution of the energy burden registered in some countries. Based on, though scarce, previous evidence, the share of households facing an energy burden higher than the 10% threshold decreased from 26% to 17% between 2013 and 2019, in Germany (Descher & Janzen, 2021). Energy affordability also seems to have improved significantly in France, after 2012, when 17% of families faced an EAR above 10% (Deller, 2018), about the same level observed in 2006 (Legendre and Ricci, 2015). Past evidence for Ireland shows that, in 2009–2010, 13% of families had affordability problems (Deller, 2018).

Despite these limitations, the analysis developed in this work provides a wider approach to energy affordability than most previous studies, particularly a deeper inequality assessment, which might be useful in future works once new data become available.

5. Conclusion

The main aim of this paper is to assess the burden brought by households' energy use in European countries, focusing on its distributive effects, which is only possible with microeconomic data.

Another objective is to provide a holistic assessment of energy affordability across countries, to uncover potential issues not revealed by a single indicator.

The magnitude and severity of affordability issues vary significantly across countries. The highest energy affordability ratios were found in countries whose average income and energy expenditure are both below the sample averages, meaning that high burdens are, in several cases, mainly explained by low income. By using a range of methods to deepen the assessment of the energy burden, our findings also showed heterogeneity across households, within countries. In the bottom two income quintiles, households face the greatest affordability issues, with concentration index findings confirming that problems are disproportionately concentrated among poorer households, even if with some nuances across countries.

The analysis performed with different metrics revealed great diversity across countries, for each indicator used, as well as varied combinations of affordability problems. These results highlight the relevance of combining several indicators to avoid neglecting the true scale of issues. Beyond revealing the existence of non-negligible affordability problems in Europe, the diagnosis developed in this work raises additional concerns due to the inflationary pressures and, particularly, the energy supply shocks that European countries have witnessed recently. However, the extent and depth of the problems are diverse across countries and also within countries. In some, serious issues are exposed by the several metrics used, while in others, major problems are revealed by a single or a few facets of affordability. Consequently, policy measures to tackle them should be tailored considering countries' idiosyncrasies, both in terms of which type of problems are more prevalent and the scope of policies: either universal or targeted at specific groups of the population.

For Bulgaria and Greece, presenting worrying results in all indicators, policies have to involve measures on several fronts, thus requiring a huge financial effort. In addition, especially in Bulgaria, it is difficult to identify families in need of support, due to the high prevalence of households struggling to pay energy bills. In this country, the only measure in place consists of cash benefits, provided in the form of means-tested heating allowances for specific seasonal periods. Thus, measures to reduce energy expenditure, whether through reduced tariffs or other ways of supporting energy consumption, are necessary and all the more urgent the more intense the problems faced by those with high ratios. However, there is a strong risk that the measures may only have palliative effects, since one of the causes for the problems identified is low income.

Curiously, the urgency of providing support was also identified for other countries, such as France, with low ratios but where the problems are highly pronounced for the few who struggle to pay energy bills. In this case, measures are easier to design and implement, for example in articulation with social security systems, to target those really need support to relieve energy bills.

In the case of Czechia, Slovakia, Estonia, Poland, Croatia and Romania, which stand out as having problematic ratios and widespread affordability issues, the policy response to mitigate the problems is also highly demanding in terms of targeting and of financially supporting the many families in need. But at least they do not perform so badly on the other indicators, which means that the effort should be directed at supporting the reduction of the energy burden, for example by offering reduced tariffs, which is only in place in Poland. The most usual measure in these countries is cash benefits.

Another group of countries for which it is difficult to design and implement affordability measures in an effective way, is that of Italy, Hungary, Latvia, Slovenia and Lithuania, due to the high shares of households with problematic ratios in combination with great prevalence of thermal discomfort and arrears. It seems that there might be some insufficiency or inadequacy of support measures in these countries; e.g., Italy already offers reduced tariffs to facilitate access to low-income households to energy services, while the remaining countries offer cash and in-kind benefits. It is necessary to maintain measures or adopt new ones to directly support consumption, such as energy vouchers, altogether with support in the form of reduced tariffs or subsidies to help households facing energy expenditures.

Both in this last group of countries and in Greece and Bulgaria, the share of households with arrears is high, but curiously only in Greece and Slovenia some limited protection against disconnections from energy supply is already in place. A good practice, learned from Hungary, that could be generalised to avoid service cuts without increasing arrears would be to provide households struggling to pay with information on existing support measures and their eligibility criteria.

In some countries, such as Portugal and Cyprus, the dichotomy between the results of the ratios and thermal comfort raises concerns about underconsumption. If we recall that no indicator, like perceived inability to keep home cool during summer time was available, added to the expected negative consequences of the climate crisis, problems in these countries may be even more significant and with a tendency to worsen. Not only in these two countries but in Southern European countries in general, the ‘inability to keep home warm’ is a real problem. Thus, it is important to design and implement solutions towards supporting consumption, complemented with the promotion of energy efficiency measures.

The results obtained from the various indicators used, in addition to revealing facets of the problem that might go unnoticed with the use of a single metric, also provide important policy implications at the Community level. For example, on the thermal efficiency of the housing stock, where problems are more similar among countries, a common policy framework is needed to mobilize EU funds to enable comprehensive and effective action. The ‘2019 Clean Energy for all Europeans Package’ is such an example in this domain, which specifically targets energy efficiency and the energy performance of buildings. But, to succeed, it is essential to overcome problems of literacy and financial incapacity that prevent those most in need from benefiting from intervention and recovery plans aimed at improving the efficiency of buildings. This objective is particularly demanding but extremely necessary, given the problems of energy poverty in Europe, which are expected to worsen due to the effects of climate change that hit more severely the most vulnerable. Our study thus highlights the need for EU co-financing of policy measures to address affordability problems, especially in countries where their prevalence and intensity is highest.

Future analyses might expand our work, analysing inequality in indicators other than the energy burden, and complementing it with research into the determinants of energy affordability problems.

Funding

This work has been supported by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., Projects UIDB/05037/2020 and UIDP/05037/2020.

The funding source was not involved in study design; collection, analysis and interpretation of data; in the writing of the report; and in the decision to submit the article for publication.

CRediT authorship contribution statement

Micaela Antunes: Methodology, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Carla Teotónio:** Conceptualization, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Carlota Quintal:** Methodology, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Rita Martins:** Conceptualization, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

None.

References

Agbim, C., Araya, F., Faust, K.M., Harmon, D., 2020. Subjective versus objective energy burden: a look at drivers of different metrics and regional variation of energy poor populations. Energy Policy 144. https://doi.org/10.1016/j.enpol.2020.111616.

Bagnoli, L., Bertoméu-Sánchez, S., 2022. How effective has the electricity social rate been in reducing energy poverty in Spain? Energy Econ. 106 https://doi.org/10.1016/j.eneco.2021.105792.

Baptista, I., Marlier, E., 2020. Access to essential services for people on low incomes in Europe. In: An Analysis of Policies in 35 Countries. European Commission, European Social Policy Network (ESPN), Brussels.

Berry, A., 2018. Measuring Energy Poverty: Uncovering the Multiple Dimensions of Energy Poverty. 2018. Hal-01896838.

Boardman, B., 1991. Fuel Poverty: From Cold Homes to Affordable Warmth. Belhaven Press, London, UK.

Bohr, J., McCreery, A., 2019. Do energy burdens contribute to economic poverty in the United States? A panel analysis. Soc. Forces. https://doi.org/10.1093/sf/soz131. November 16.

Brown, M.A., Soni, A., Lapsa, M.V., Southworth, K., Cox, M., 2020. High energy burden and low-income energy affordability: conclusions from a literature review. Prog. Energy 2 (4), 042003. https://doi.org/10.1088/2516-1083/abb954.

Castaño-Rosa, R., Solís-Guzmán, J., Rubio-Bellido, C., Marrero, M., 2019. Towards a multiple-indicator approach to energy poverty in the European Union: a review. Energ. Build. 193, 36–48. https://doi.org/10.1016/j.enbuild.2019.03.039.

CEER, 2014. The implementation of the 2020 vision for Europe’s energy customers by its supporters. In: CEER Report C14-SC-20-03. Council of European Energy Regulators, Brussels.

Colton, R.D., 2011. Home Energy Affordability in New York: The Affordability Gap (2008–2010) Prepared for New York State Energy Research Development Authority (NYSERDA) Albany, New York.

Cook, J.J., Shah, M., 2018. Reducing Energy Burden with Solar : Colorado ’ s Strategy and Roadmap for States Reducing Energy Burden with Solar. March, 1–3. https://www.nrel.gov/docs/fy18osti/70965.pdf.

Das, R.R., Martiskainen, M., Li, G., 2022. Quantifying the prevalence of energy poverty across Canada: Estimating domestic energy burden using an expenditures approach. In: Canadian Geographer. https://doi.org/10.1111/cag.12750.

Davillas, A., Burlinson, A., Liu, H.-H., 2022. Getting warmer: fuel poverty, objective and subjective health and well-being. Energy Econ. 106 https://doi.org/10.1016/j.eneco.2021.105794.

Deller, D., 2018. Energy affordability in the EU: the risks of metric driven policies. Energy Policy 119 (December 2017), 168–182. https://doi.org/10.1016/j.enpol.2018.03.033.
Deller, D., Waddams, C., 2015. Affordability of Utilities' Service: extent, practice and policy, research paper 1 – affordability metrics. In: Centre of Regulation on Europe. 22 October 2015. Available at. https://cerre.eu/wp-content/uploads/2020/07/151022_CERRE_AffordabilityUtilitiesServices_ResearchPaper_1.pdf.
Deller, D., Turner, G., Waddams Price, C., 2021. Energy poverty indicators: inconsistencies, implications and where next? Energy Econ. 103 (September), 105551. https://doi.org/10.1016/j.eneco.2021.105551.
Dobbins, A., Nerini, F.F., Pye, S., 2016. Measures to Protect Vulnerable Consumers in the Energy Sector: An Assessment of Disconnection Safeguards, Social Tariffs and Financial Transfers, INSIGHT-E Programme, Policy Report, December 2016. European Commission. http://bit.do/eM4kf.
Drehobl, A., Ross, L., Ayala, R., 2020. How High Are Household Energy Burdens? American Council for an Energy-Efficient Economy, Washington, DC.
Drescher, K., Janzen, B., 2021. Determinants, persistence, and dynamics of energy poverty: an empirical assessment using German household survey data. Energy Econ. 102 (January), 105433. https://doi.org/10.1016/j.eneco.2021.105433.
Dubois, U., Meier, H., 2016. Energy affordability and energy inequality in Europe: implications for policymaking. Energy Res. Soc. Sci. 18, 21–35. https://doi.org/10.1016/j.erss.2016.04.015.
EEA, 2017. Climate change, impacts and vulnerability in Europe 2016 an indicator-based report. In: European Environment Agency. Luxembourg: Publications Office of the European Union. https://doi.org/10.2800/534806.
EPOV, 2022. EU Energy Poverty Observatory. European Commission. https://energy-poverty.ec.europa.eu/ (accessed 19th May 2022).
EU, 2019a. Directive (EU) 2019/692 of the European Parliament and of the Council of 17 April 2019 Amending Directive 2009/73/EC Concerning Common Rules for the Internal Market in Natural Gas (OJ L 117, 3.5.2019, p. 1–7).
EU, 2019b. Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on Common Rules for the Internal Market for Electricity and Amending Directive 2012/27/EU (OJ L 158, 14.6.2019, p. 125–199).
European Commission, 2019. Reflection paper: Towards a Sustainable Europe by 2030. https://www.eesc.europa.eu/en/our-work/opinions-information-reports/opinions/reflection-paper-towards-sustainable-europe-2030.
European Commission, Directorate-General for Employment, Social Affairs and Inclusion, Baptista, I., Marlier, E., 2020. Access to essential services for people on low incomes in Europe: an analysis of policies in 35 countries: 2020. In: Publications Office, 2020. https://doi.org/10.2767/93987.
European Union, 2021. Sustainable Development in the European Union - Monitoring Report on Progress towards the SDGs in an EU Context. Publications Office of the European Union, Luxembourg. https://doi.org/10.9774/gleaf.9781315611471.
Eurostat, 2022a. Eurostat Database – Population Unable to Keep Home Adequately Warm by Poverty Status. Available at. https://ec.europa.eu/eurostat/databrowser/view/sdg_07_60/default/table?lang=en (accessed 4th May 2022).
Eurostat, 2022b. Eurostat Database – Population with Arrears on Utility Bills. Available at. https://ec.europa.eu/eurostat/databrowser/view/ilc_mdes07/default/table?lang=en (accessed 4th May 2022).
Eurostat, 2022c. Eurostat Database – Living in a Dwelling with a Leaking Roof, Damp Walls, Floors or Foundation or Rot in Window Frames of Floor by Poverty Status. Available at. https://ec.europa.eu/eurostat/databrowser/view/sdg_01_60/default/table?lang=en (accessed 4th May 2022).
Fankhauser, S., Tepic, S., 2007. Can poor consumers pay for energy and water? An affordability analysis for transition countries. Energy Policy 35 (2), 1038–1049. https://doi.org/10.1016/j.enpol.2006.02.003.
Fournier, E.D., Cudd, R., Federico, F., Pincetl, S., 2020. On energy sufficiency and the need for new policies to combat growing inequities in the residential energy sector. Elementa: Sci. Anthropocene 8.
Frondel, M., Sommer, S., Vance, C., 2015. The burden of Germany's energy transition: an empirical analysis of distributional effects. Econ. Anal. Policy 45 (2015), 89–99. https://doi.org/10.1016/j.eap.2015.01.004.
Gawel, E., Bretschneider, W., 2014. The affordability of water and energy pricing. In: Fitzpatrick, T. (Ed.), International Handbook on Social Policy and the Environment (Pp. 123–151). Edward Elgar. https://doi.org/10.4337/9780857936134.
Goddard, J.J., Ray, I., Balazs, C., 2021. Water affordability and human right to water implications in California. PLoS One 16 (1 January), 3–5. https://doi.org/10.1371/journal.pone.0245237.
Gouveia, J.P., Palma, P., Bessa, S., Mahoney, K., Sequeira, M., 2022. Energy Poverty National Indicators. Insights for a More Effective Measuring. Energy Poverty Advisory Hub. European Commission.
Hills, J., 2012. Getting the Measure of Fuel Poverty Final Report of the Fuel Poverty Review. Centre for Analysis of Social Exclusion. London School of Economics and Political Science, London.
Isherwood, B.C., Hancock, R.M., 1979. Household Expenditure on Fuel: Distributional Aspects. Economic Advisor's Office, Department for Health and Social Security, London.
Kessides, I., Miniaci, R., Scarpa, C., Valbonesi, P., 2009. Towards Defining and Measuring the Affordability of the Public Utility Services, Policy Research Working Paper 4915. The World Bank, Development Research Group, Environment and Energy Team.
Kyprianou, I., Serghides, D.K., Varo, A., Gouveia, J.P., Kopeva, D., Murauskaite, L., 2019. Energy poverty policies and measures in 5 EU countries: a comparative study. Energ. Build. 196, 46–60. https://doi.org/10.1016/j.enbuild.2019.05.003.
Legendre, B., Ricci, O., 2015. Measuring fuel poverty in France: which households are the most fuel vulnerable? Energy Econ. 49, 620–628. https://doi.org/10.1016/j.eneco.2015.01.022.
Lin, J., 2018. Affordability and access in focus: metrics and tools of relative energy vulnerability. Electr. J. 31 (6), 23–32. https://doi.org/10.1016/j.tej.2018.06.005.
Maricopa County Department of Public Health Office of Epidemiology, 2016. Assessing the Cooling Needs of Homebound Individuals in Maricopa County, 2016. Phoenix: Maricopa County Department of Public Health.
Martins, R., Quintal, C., Antunes, M., 2019. Making ends meet: actual versus potential joint affordability of utility services. Util. Policy 56 (December 2018), 120–126. https://doi.org/10.1016/j.jup.2018.12.002.
Meier, H., Jamasb, T., Orea, L., 2013. Necessity or luxury good? Household energy spending and income in Britain 1991–2007. Energy J. 34 (4), 109–128.
Moore, R., 2012. Definitions of fuel poverty: implications for policy. Energy Policy 49, 19–26. https://doi.org/10.1016/j.enpol.2012.01.057.
Mulder, P., Dalla Longa, F., Straver, K., 2023. Energy poverty in the Netherlands at the national and local level: a multi-dimensional spatial analysis. Energy Res. Soc. Sci. 96, 102892. https://doi.org/10.1016/j.erss.2022.102892.
O'Donnell, O., van Doorslaer, E., Wagstaff, A., Lindelow, M., 2008. Analyzing Health Equity Using Survey Data: A Guide to Techniques and their Implementation. The World Bank, Washington, DC.
O'Donnell, O., O'Neill, S., Van Ourti, T., Walsh, B., 2016. *Conindex:* estimation of concentration indices. Stata J. 16 (1), 112–138.
OECD, 2018. Measuring energy poverty and ways to protect vulnerable groups in the European Union. In: Energy Subsidy Reform in the Republic of Moldova: Energy Affordability, Fiscal and Environmental Impacts. OECD Publishing, Paris. https://doi.org/10.1787/9789264292833-5-en.
Oliveira-Panão, M.J.N., 2021. Lessons learnt from using energy poverty expenditure-based indicators in a mild winter climate. Energy Build. 242. https://doi.org/10.1016/j.enbuild.2021.110936.
Richardson, P., 1978. Fuel Poverty: A Study of Fuel Poverty among Low Income Council Tenants. Department of Social Administration and Social Work, University of York, York.
Romero, J.C., Linares, P., López, X., 2018. The policy implications of energy poverty indicators. Energy Policy 115, 98–108. https://doi.org/10.1016/j.enpol.2017.12.054.
Rowley, S., Ong, R., Haffner, M., 2015. Bridging the gap between housing stress and financial stress: the case of Australia. Hous. Stud. 30 (3), 473–490. https://doi.org/10.1080/02673037.2014.977851.
Simshauser, P., 2021. Vulnerable households and fuel poverty: measuring the efficiency of policy targeting in Queensland. Energy Econ. 101 (February), 105405. https://doi.org/10.1016/j.eneco.2021.105405.
Stigler, G.J., 1954. The early history of empirical studies of consumer behavior. J. Polit. Econ. 62 (2), 95–113. http://www.jstor.org/stable/1825569.
Stiglitz, J., 2013. The Price of Inequality: How Today's Divided Society Endangers our Future. New York.
Taylor, L., 1993. Fuel poverty: from cold homes to affordable warmth. Energy Policy 21 (10), 1071–1072. https://doi.org/10.1016/s0301-4215(06)80013-3.
Teotónio, C., Martins, R., Antunes, M., Quintal, C., 2023. Unveiling underconsumption of water and electricity services at the bottom of the income distribution. Util. Policy 82, 101572. https://doi.org/10.1016/j.jup.2023.101572.
Thomson, H., Bouzarovski, S., 2018. Addressing Energy Poverty in the European Union: State of Play and Action. Report for EU Energy Poverty Observatory, ENER/B3/SER/2015–507/SI2.742529.
Thomson, H., Bouzarovski, S., Snell, C., 2017. Rethinking the measurement of energy poverty in Europe: a critical analysis of indicators and data. Indoor Built Environ. 26 (7), 879–901. https://doi.org/10.1177/1420326X17699260.
Tirado-Herrero, S., 2017. Energy poverty indicators: a critical review of methods. Indoor Built Environ. 26 (7), 1018–1031. https://doi.org/10.1177/1420326X17718054.
Trinomics, 2016. Selecting Indicators to Measure Energy Poverty – Final Report. Report for DG-Energy, ENER/B3/507–2015, Rotterdam.
Yoon, H., Saurí, D., 2019. 'No more thirst, cold, or darkness!' – Social movements, households, and the coproduction of knowledge on water and energy vulnerability in Barcelona, Spain. Energy Res. Soc. Sci. 58 (September 2018), 101276. https://doi.org/10.1016/j.erss.2019.101276.